Virtual University of Pakistan

BIO204

Biochemical Engineering

Final Terms Past Papers

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Group of VU Biologists

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Bio204

Requirements to run a fermenter..
Batch and continuous..
Parts of fermenter..
How sterilized product exhaust air.. Aisa he tha..
Richard sterilization cycle..
Chemostat
Long 10 marks
Oxygen delivery in foam control system..
Recovery and purification..
Factors consider when designing a fermenter..

BIO204 Final term paper
1.Write any Two Risks described by Collins? 2
2. What is Sterilization? 2
3. Requirements to run a fermenter? 2
4. How sterilized product exhaust air? 3
5. Factors consider when designing a fermenter? 5
6. Richards' RAPID METHOD FOR DESIGNING OF STERILIZATION CYCLES? 5
7. Discuss Oxygen delivery system and Form control system? 10
8. Discuss batch and continuous sterlization process, filter Sterilization of Media, Sterilization of the Fermenter, Feeds and of Liquid Wastes. 10

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Mcqs
In Xt = **xo** eμt, **Xo** what is **xo** and xt in this equation
Nitryl or butyl rubbers are normally used for these (seals.) ya mcq tha
11.If qp is strictly growth related then it will change as μ with
D and, thus, the product concentration remain (constant.)
Baqi b mid k pprs sy thy kuch quiz sy thy kuch bongy thy .....
Questions:
Components of fermenter
What is chemostate(2)
product recovry 3mrks ,
sterilization 2 mrks ,
del factor equation, 3 mrks
fermenr parts 3 mrks
requirements for fermentation 5 mrks

Roffler et al. (1984) techniques 5 mrks
Effluent treatment 5mrks
how to sterilize air exhaust 2 mrks

richrd wala tha ak 3 mrks
Relatio of qp and M (mew) in fedbatch culture (3)

Bio204 ppr today's ppr
What is sterilization?
How does sterilize the fermenter exhaust air?
What is batch and continuous culture?
Richards method of integration?
Parts of fermenter?
Factors required for fermentation?
Write the equation of overall del factor?

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All paper from past expect 2
Explain d and y from equation
Equation yd ni
Nd
Batch nd continues culture in detail

Bio204 (20-08-2017)
Total Q 47
34 Mcqs
Short Q:
1: What is Sterlization? 2 marks
2: What is the parts of fermenter? 2 marks
3: What are the requirement to run fermenter? 3 marks
4: How sterlized the exhust air? 3 marks
5: Richards rapid method for designing of sterlization cycles? 2 marks
6: What is the first stage recovery of fermentation? 2 marks
7: Del equation? 2 marks
8: Xt/x0=e –kt 3 marks
9:industrial waste treatment syatem factore? 3 marks
10. Characteristics of fermenter seal? 2 marks
Long Q
11: Batch fermentation and contineou fermentation? 5 marks
12 Techniques for the fermentation of in-situ fermentation products recover by Roffler et al 1984? 5 marks
13: factors consider when designing a fermenter? 5 marks